VEIN: On-Chain Causal Risk Attribution for Cryptocurrency Systemic Risk

Quang-Vinh Dang[1*]

[1*]British University Vietnam, Hung Yen, Vietnam.

Corresponding author(s). E-mail(s): vinh.dq4@buv.edu.vn;

Abstract

Existing causal measures of systemic risk in cryptocurrency markets infer the contagion graph *statistically* from price returns, using constraint-based structure learning or directed-information estimation. We propose VEIN (Verifiable Entity-resolved Interventional/counterfactual Network risk), a framework that instead treats the *observed* on-chain transaction graph as the structural backbone of a Pearl-style structural causal model (SCM): the causal parent set of each economic entity is read off the ledger rather than estimated from correlations. On this graph VEIN defines an interventional risk measure (On-Chain Causal CoVaR) through the do-operator and, to our knowledge, the first counterfactual (Pearl Level-3) systemic-risk attribution measure in finance, which decomposes a realised loss into the portion attributable to a specific upstream entity's distress. We implement the complete pipeline—a four-tier entity-resolution layer over Dune Analytics' decoded Ethereum tables, the SCM, and the risk measures—and evaluate it on real on-chain data around the October 2025 cascading liquidation event, the largest deleveraging episode in crypto history. At the event's native (hourly) timescale, the observed graph matches or slightly outperforms a price-estimated Granger graph at predicting the cascade; the OC-CoVaR systemic ranking diverges from $\boldsymbol{\Delta}$CoVaR and MES (Spearman $\boldsymbol{\rho = 0.52}$); the counterfactual measure attributes concrete loss shares between exchanges; and a battery of causal-validity checks (placebo/negative controls, unobserved-confounding robustness values, resolution robustness, and external validity against the documented event narrative) corroborates the framework. The one assumption that does *not* survive falsification is that flow *direction* encodes causal direction: edge reversal barely changes predictions, indicating that causal information resides in the network's connectivity structure rather than in flow orientation. We discuss the implications for regulators and the limits imposed by centralised-exchange opacity. JEL classification: G01, G17, G18, C58.

Keywords: systemic risk, cryptocurrency, structural causal models, counterfactual inference, blockchain, CoVaR, contagion

1 Introduction

On 10–11 October 2025, cryptocurrency markets experienced the largest deleveraging event in their history: on the order of nineteen billion US dollars of leveraged positions were liquidated within thirty-six hours.[1] Unlike the 2022 episodes (Terra/Luna, Celsius, FTX), which were rooted in fundamental insolvency, the October 2025 cascade was largely *mechanical*: a mispricing in a centralised exchange's internal valuation engine[2] propagated into automated liquidations, which in turn forced sales that depressed prices and triggered further liquidations—a classic funding-liquidity spiral operating at machine speed. A defining feature was architectural: decentralised finance (DeFi) lending protocols with transparent, over-collateralised positions (e.g. Aave) emerged with essentially zero bad debt, whereas centralised-exchange (CEX) risk engines amplified the shock. The synchronised depeg of the synthetic dollar USDe—which remained fully solvent throughout, so that the depeg was a pricing and liquidity artefact rather than a default—is emblematic of this architecture-dependent fragility.[3]

[1] For event overviews and the minute-level liquidation mechanics, see CoinDesk Research, https://www.coindesk.com/research/market-spotlight-the-19-billion-liquidation-that-shook-crypto; CoinGecko, https://www.coingecko.com/learn/october-10-crypto-crash-explained; and FTI Consulting, https://www.fticonsulting.com/insights/articles/crypto-crash-october-2025-leverage-met-liquidity.

[2] The proximate trigger has been attributed to Binance's unified-account margin system, which valued posted collateral (including USDe, wBETH and BNSOL) using its own internal spot prices rather than external oracles; see https://www.ccn.com/education/crypto/ethena-usde-depeg-binance-crash-explained/ and https://ambcrypto.com/not-a-true-depeg-says-ethena-founder-as-usde-faces-binance-only-slip/.

[3] USDe's price fell to roughly \$0.65 on Binance while holding near \$1 on other venues and in DeFi, and remained redeemable as supply contracted from about \$9bn to \$6bn, confirming an exchange-specific pricing artefact rather than insolvency; see CoinDesk, https://www.coindesk.com/markets/2025/10/11/ethena-s-usde-briefly-loses-peg-during-usd19b-crypto-liquidation-cascade and https://www.coindesk.com/markets/2025/10/13/no-ethena-s-usde-didn-t-de-peg.

Crucially for risk measurement, the proximate mechanism left a high-resolution footprint *on chain*: liquidation transfers, collateral movements, and exchange in/out-flows are all recorded, timestamped to the block, in a public ledger.

Such episodes expose a structural weakness in how systemic risk is measured for crypto markets. The dominant tools—ΔCoVaR [1], marginal expected shortfall and SRISK [2], and variance-decomposition connectedness [3, 4]—are *associational*: they quantify co-movement in returns and, as their own authors note, capture an institution's contribution to system risk "in a non-causal sense" [1]. A more recent literature attempts to inject causality. Causal-NECO VaR [5, 6] learns a contagion graph from foreign-exchange returns using the order-independent PC-stable algorithm [7, 8] and computes a value-at-risk from the resulting SCM. The time-varying directed-information graph (TV-DIG) of Etesami et al. [9] builds directed networks—an information-theoretic generalisation of Granger causality [10]— across financial sectors including cryptocurrencies. Correlation-network models such as CoRisk [11] decompose systemic risk into idiosyncratic and contagion components. Yet every one of these methods shares a common limitation: *the causal graph itself is estimated from prices*, so structure-learning error and data error compound, and no published method exploits the one data source where the contagion mechanism is *directly observed*—the blockchain ledger.

Separately, on-chain transaction graphs have been used for prediction and forensics—"chainlets" that forecast Bitcoin price and volatility [12], address-clustering for anti-money-laundering [13, 14]—but always observationally, explicitly disclaiming a causal-structural interpretation, and never as the backbone of a systemic-risk measure. And although Pearl's causal hierarchy [15, 16] places counterfactuals (Level 3) above interventions (Level 2) above association (Level 1), no systemic-risk measure in finance operates above Level 2. There is, in short, an open intersection: *an observed-graph SCM for crypto systemic risk, extended to the counterfactual level.*

Contribution. This paper makes three contributions.

1. **Observed graph as SCM.** We propose VEIN, which treats the entity-resolved on-chain transaction graph as the structural backbone of a Pearl SCM. The causal parent set $\mathrm{pa}(i)$ of each entity is *read off the ledger*, not inferred from price correlations. The temporal precedence of on-chain transactions (guaranteed by block timestamps) supplies an identification advantage unavailable to price-based methods.
2. **Counterfactual systemic risk.** We define On-Chain Causal CoVaR, an interventional (Level-2) measure via the do-operator, and a counterfactual (Level-3) attribution measure that decomposes a realised loss into the share attributable to a specific upstream entity, via the standard abduction–action–prediction procedure.
3. **End-to-end implementation and honest evaluation.** We implement the full pipeline—including a four-tier entity-resolution layer over real decoded Ethereum data—and evaluate it on the October 2025 event, comparing against price-estimated graphs and standard non-causal baselines, and subjecting every causal claim to a falsification and validity battery. We report negative results as readily as positive ones.

Hypotheses. We organise the evaluation around five hypotheses (Table 1), each paired in Section 5 with the specific test and falsification criterion that would overturn it.

The remainder of the paper is organised as follows. Section 2 positions VEIN against the causal and on-chain literatures. Section 3 develops the framework formally. Section 4 describes the data and the entity-resolution pipeline. Section 5 reports the empirical evaluation and validity battery. Section 6 discusses implications, limitations, and conclusions. An appendix collects implementation details and additional tables.

Table 1 Hypotheses evaluated in this paper.

ID	Statement
H1	The observed on-chain flow graph predicts stress propagation more accurately than a graph statistically estimated from price returns.
H2	Reversing flow-edge direction degrades predictive accuracy (flow direction carries causal information; assumption A3).
H3	OC-CoVaR identifies a different set of systemically important entities than ΔCoVaR/MES applied to returns.
H4	The counterfactual measure decomposes the October 2025 losses into mechanically interpretable components.
H5	On-chain composability/collateral edges add loss-attribution information beyond price correlation.

2 Related Work and Positioning

Associational systemic-risk measures. The canonical conditional measure is ΔCoVaR [1], the change in the value-at-risk of the financial system conditional on an institution being in distress relative to its median state. The systemic-expected-shortfall family [2]—marginal expected shortfall (MES) and the leverage-adjusted SRISK—measures an institution's losses in the tail of the system distribution. Connectedness measures based on forecast-error-variance decompositions [3] and on Granger-causal and correlation networks [4] quantify spillovers. All are explicitly non-causal: they describe co-movement, not mechanism.

Causal value-at-risk. The closest methodological precedent is Causal-NECO VaR [5], which decomposes asset log-returns into an autoregressive part and an instantaneous contagion part, learns the causal structure with PC-stable [7], and computes a Gaussian or Gaussian-copula VaR. It operates at Pearl Levels 1–2 on foreign-exchange data, assumes no unmeasured confounders, and—decisively for our purposes—obtains its graph by estimation rather than observation. The foundational lineage [6] combines causal networks with a structural vector autoregression, drawing on Pearl's framework and the PC algorithm [8]. VEIN differs on all three axes: an observed on-chain graph, crypto entities, and a counterfactual (Level-3) extension.

Causal systemic-risk networks including crypto. TV-DIG [9] is the nearest crypto application: a nonparametric, time-varying directed-information graph that detects heightened crypto-to-sector influence around the 2022 stress events. Directed information generalises Granger causality [10] but is not a Pearl SCM with do-operator/counterfactual semantics, and it is built from returns, not on-chain flows.

On-chain graphs and entity resolution. Blockchain transaction graphs have been mined predictively—chainlet subgraphs forecasting Bitcoin price and volatility [12]—and for forensics and address clustering, from the multi-input/co-spend heuristics of [13] to graph-convolutional anti-money-laundering classifiers on the Elliptic dataset [14]. These establish the resolution machinery VEIN reuses, but none treats the resulting graph as a causal structure for a systemic-risk measure.

The gap and our position. Two specific capabilities are absent from the published literature as of mid-2026: (i) a counterfactual (Level-3) VaR/CoVaR measure, and (ii) the use of an observed on-chain transaction graph as a Pearl SCM for systemic risk. Table 2 situates VEIN against the nearest neighbours. We position our contribution against Causal-NECO VaR (nearest method) and TV-DIG (nearest crypto application), and re-implement the "estimated-graph" approach on our own data as the head-to-head baseline for H1.

3 The VEIN Framework

3.1 Notation and setup

Let $G = (V, E)$ be a directed, timestamped, dollar-weighted graph whose nodes V are *resolved economic entities* and whose edges E are observed on-chain flows. For each entity i and discrete time t we define a scalar *stress state* $S_{i,t} \in \mathbb{R}$ (oriented so that

Table 2 Positioning of VEIN against the nearest prior art.

Method	Graph source	Causal level	Domain	Counterfactual
ΔCoVaR [1]	none (returns)	assoc. (L1)	general	no
MES/SRISK [2]	none (returns)	assoc. (L1)	general	no
CoRisk [11]	corr. network	assoc. (L1)	credit	no
TV-DIG [9]	estimated (DI)	Granger	incl. crypto	no
Causal-NECO VaR [5]	estimated (PC)	SCM (L1–2)	FX	no
VEIN (this paper)	**observed**	**SCM (L1–3)**	**crypto**	**yes**

Table 3 Notation.

Symbol	Meaning
$G = (V, E)$	resolved entity graph (nodes, directed flow edges)
$\mathrm{pa}(i)$	causal parents of i (in-neighbours in G)
$S_{i,t}$	stress state of entity i at time t (z-scored)
$F_{p \to i,t}$	observed USD flow from p to i in period t
$U_{i,t}$	exogenous shock to i at t
f_i	structural (stress-propagation) function of i
L_j	loss functional of j over the event window
s^*	severe-distress intervention level (95th pctile)
$\Delta_i^{CF}(j)$	counterfactual loss of j attributable to i
RV	confounding robustness value

larger means more distressed), the realised flow $F_{p \to i,t} \geq 0$ from p to i during period t, and an exogenous shock $U_{i,t}$. Table 3 summarises the notation.

3.2 Structural causal model

The defining structural equation of VEIN is

$$S_{i,t} \;=\; f_i\Big(S_{\mathrm{pa}(i),\,t-1},\; F_{\mathrm{pa}(i) \to i,\,t},\; U_{i,t}\Big), \tag{1}$$

where the parent set

$$\mathrm{pa}(i) \;=\; \{\, p \in V : (p \to i) \in E \,\} \tag{2}$$

is *read directly from the resolved ledger.* This is the central structural move of the paper: in contrast to [5, 9], $\mathrm{pa}(i)$ is not estimated from data but observed. The only component requiring statistical estimation is the stress-propagation function f_i. We adopt a ridge-regularised linear specification,

$$S_{i,t} \;=\; b_i + \sum_{p\in\mathrm{pa}(i)} a_{ip}\, S_{p,t-1} + \sum_{p\in\mathrm{pa}(i)} c_{ip}\, \log\bigl(1 + F_{p\to i,t}\bigr) + U_{i,t}, \tag{3}$$

with parameter vector $\theta_i = (b_i, \{a_{ip}\}, \{c_{ip}\})$ estimated by ridge regression,

$$\hat{\theta}_i \;=\; \arg\min_{\theta_i} \sum_t \bigl(S_{i,t} - f_i(\cdot;\theta_i)\bigr)^2 + \lambda\bigl(\|a_{i\cdot}\|_2^2 + \|c_{i\cdot}\|_2^2\bigr), \tag{4}$$

on a pre-crisis estimation window, with penalty λ. The fitted residuals $\widehat{U}_{i,t} = S_{i,t} - f_i(\cdot;\hat{\theta}_i)$ furnish an empirical distribution of exogenous shocks used below.

Proposition 1 (Well-posedness of the lagged system) *Because every parent enters* (3) *at lag* t–1 *in the stress term and the contemporaneous term depends only on exogenous observed flows* $F_{p\to i,t}$*, the map from* $(S_{\cdot,t-1}, F_{\cdot,t}, U_{\cdot,t})$ *to* $S_{\cdot,t}$ *is explicit. Hence, given an initial state* $S_{\cdot,0}$ *and a shock path* $\{U_{\cdot,t}\}$*, the trajectory* $\{S_{\cdot,t}\}_{t\geq 1}$ *is uniquely determined by forward iteration, even when* G *contains directed cycles.*

Proof sketch At step t the right-hand side of (3) for every i references only quantities indexed at $t-1$ (parent stress) or exogenous quantities at t (flows, shocks). There is therefore no simultaneity among the time-t stress variables, and $S_{\cdot,t}$ is computed in one vectorised step from information available at $t-1$. Induction on t gives existence and uniqueness. □

Proposition 1 is what allows cycles such as Binance↔retail in the observed graph without an equilibrium fixed-point computation: time, supplied by the ledger, breaks the simultaneity.

3.3 Identifying assumptions

Assumption 1 (Temporal precedence, A1) A flow $F_{i \to j,t}$ strictly precedes any effect on j at $t' > t$.

A1 is supplied *for free* by blockchain immutability and timestamping; it need not be assumed or tested, unlike Granger-style precedence on price series.

Assumption 2 (No unobserved common cause for flow edges, A2) Conditional on the transaction graph up to t, an edge $i \to j$ reflects i's direct economic action on j.

A2 is defensible because a public ledger captures every fund movement in the flow mechanism itself. Its principal threat is off-chain activity—most acutely CEX-internal pricing engines—which we bound through a sensitivity analysis (Section 3.7).

Assumption 3 (Monotone propagation along flow direction, A3) Distress at i affects j only through realised flow paths from i to j.

A3 is the substantive, falsifiable claim of the framework and is tested directly (H2) by the edge-reversal experiment of Section 3.6. A1–A2 are structural properties of the data source—the principal identification advantage over price-based causal discovery, where both the graph and the equations must be estimated, compounding error.

3.4 Interventional measure: On-Chain Causal CoVaR

We summarise an entity's distress over an event window $\mathcal{T}$ by the non-negative, additive loss functional

$$L_j \;=\; \sum_{t \in \mathcal{T}} \max\bigl(S_{j,t},\, 0\bigr). \tag{5}$$

Applying Pearl's do-operator to (1), we force entity i into a distressed state s^*, severing its own parents, and propagate forward through the remaining structural equations.

The interventional risk measure is the conditional value-at-risk of j's loss,

$$\text{OC-CoVaR}_q\big(j \,\big|\, \mathrm{do}(S_i = s^*)\big) \;=\; \mathrm{VaR}_q\Big(L_j \,\Big|\, \mathrm{do}(S_i = s^*)\Big), \tag{6}$$

whose distribution is obtained by Monte-Carlo propagation. For $m = 1, \dots, M$ we draw a shock path $U^{(m)}$ by block-bootstrapping the fitted residuals $\{\widehat{U}_{i,t}\}$, simulate (3) forward under the intervention to obtain $S_{j,t}^{(m)}$, and form $L_j^{(m)}$ via (5); the estimator is the empirical quantile

$$\widehat{\text{OC-CoVaR}}_q\big(j \,|\, \mathrm{do}(S_i = s^*)\big) \;=\; \widehat{Q}_q\big(\{L_j^{(m)}\}_{m=1}^{M}\big). \tag{7}$$

By analogy with ΔCoVaR, the marginal systemic contribution of i to j is

$$\Delta\text{OC-CoVaR}_q(j \,|\, i) \;=\; \text{OC-CoVaR}_q\big(j \,|\, \mathrm{do}(S_i = s^*)\big) - \text{OC-CoVaR}_q\big(j \,|\, \mathrm{do}(S_i = 0)\big), \tag{8}$$

with s^* the 95th percentile of i's historical stress and the baseline the median (0 after standardisation). We rank entities by their total *exported* tail risk,

$$\mathcal{E}(i) \;=\; \sum_{j \neq i} \max\big(\Delta\text{OC-CoVaR}_q(j \,|\, i),\, 0\big). \tag{9}$$

Relation to ΔCoVaR. Equation (8) is the network-causal analogue of ΔCoVaR [1]: both contrast a distressed against a median conditioning state. The differences are structural. ΔCoVaR conditions on a *statistical* event (institution i at its VaR) and reads off a conditional quantile of returns; ΔOC-CoVaR conditions on an *intervention* ($\mathrm{do}(S_i = s^*)$) that severs i's own parents and propagates the shock forward through the observed structural equations. In the degenerate case of a single edge $i \to j$ with a linear f_j and no further descendants, ΔOC-CoVaR reduces to a quantile shift of j

proportional to the structural coefficient a_{ji}, mirroring the β_i slope in ΔCoVaR; the network generalisation is the sum over all directed paths from i to j.

3.5 Counterfactual measure: loss attribution

The counterfactual attribution of j's realised loss to i's distress follows the three-step procedure [15, 16]: *(i) Abduction*—infer the realised exogenous shocks $\widehat{U}_{i,t} = S_{i,t} - f_i(\cdot;\hat{\theta}_i)$ from observed data; *(ii) Action*—replace i's realised trajectory by its pre-crisis level, $\mathrm{do}(S_i = S_i^{\mathrm{pre}})$, holding all $\widehat{U}_{k,t}$ $(k \neq i)$ fixed; *(iii) Prediction*—recompute downstream states to obtain $L_j^{\mathrm{do}(S_i = S_i^{\mathrm{pre}})}$. The attribution and its share are

$$\Delta_i^{CF}(j) \;=\; L_j^{\mathrm{obs}} - L_j^{\mathrm{do}(S_i = S_i^{\mathrm{pre}})}, \qquad \mathrm{share}_i(j) \;=\; \frac{\Delta_i^{CF}(j)}{L_j^{\mathrm{obs}}}. \tag{10}$$

$\Delta_i^{CF}(j)$ answers: *how much of j's realised loss would not have occurred had i never become distressed, holding everything else exactly as it unfolded?* To our knowledge this is the first counterfactual (Level-3) systemic-risk attribution in finance.

3.6 Falsification test for A3

To test, rather than assume, that flow direction carries causal information we construct three variants of G: the true-direction graph, a reversed graph G^{rev} (every edge flipped), and a symmetric/undirected graph G^{sym}. We fit (3) on each over the estimation window and compare one-step-ahead prediction of event-window stress, scoring the stress-weighted root-mean-square error

$$\mathrm{RMSE} \;=\; \sqrt{\frac{\sum_{i,t} w_{i,t}\,(\hat{S}_{i,t} - S_{i,t})^2}{\sum_{i,t} w_{i,t}}}, \qquad w_{i,t} = |S_{i,t}| + 0.1. \tag{11}$$

If the true-direction model attains the lowest RMSE this is empirical evidence for A3; if it does not, A3 must be qualified to specific edge types.

3.7 Sensitivity to unobserved confounding

Because A2 is not directly testable, we quantify how strong a hidden confounder (e.g. a CEX-internal pricing engine) would have to be to overturn each edge, using the robustness value of Cinelli and Hazlett [17]. For a structural coefficient with t-statistic t and d degrees of freedom, let $f = |t|/\sqrt{d}$; the robustness value for nullifying the estimate is

$$RV \;=\; \tfrac{1}{2}\Big(\sqrt{f^4 + 4f^2} - f^2\Big) \in [0,1], \tag{12}$$

the share of residual variance an unobserved confounder must explain in *both* the parent (treatment) and the child (outcome) to drive the coefficient to zero. Larger RV means a finding is harder to explain away.

3.8 Baselines

We compare against three references. *(i)* $\Delta CoVaR$ [1]: estimating the quantile regression $r_t^{\text{sys}} = \alpha_i + \beta_i\, r_{i,t} + \varepsilon_t$ at level q, the measure is $\Delta\mathrm{CoVaR}_i = \beta_i\big(\mathrm{VaR}_q(r_i) - \mathrm{VaR}_{0.5}(r_i)\big)$. *(ii) MES* [2]: the mean return of asset i on the worst-q system days, $\mathrm{MES}_i = \mathbb{E}[r_i \mid r^{\text{sys}} \le \mathrm{VaR}_q(r^{\text{sys}})]$. *(iii) The estimated-graph school*: a Granger graph [10] (edge $i \to j$ when the lagged return of i significantly predicts that of j in $r_{j,t} = \phi_0 + \phi_1 r_{j,t-1} + \phi_2 r_{i,t-1} + u_t$ at level α) and a partial-correlation graph [11], both over the same entity node set and fed through the *same* VEIN machinery so that only the graph source differs—a like-for-like test of H1.

3.9 Algorithm

Algorithm 1 summarises the end-to-end procedure.

Algorithm 1 VEIN end-to-end pipeline

Input: decoded on-chain transfers, price/return data, entity labels, event window $\mathcal{T}$
$G \leftarrow \textsc{ResolveEntities}(\text{transfers}, \text{labels})$ ▷ Tiers 0–3, Sec. 4
for each entity $i \in V$ **do**
  $\hat{\theta}_i \leftarrow \textsc{FitRidge}\big(\text{pa}(i, G),\ F_{\text{pa}(i)\to i},\ S_i^{\text{hist}}\big)$ ▷ Eq. (4)
end for
for each (i, j) of interest **do**
  $\Delta\text{OC-CoVaR}_q(j \,|\, i) \leftarrow \textsc{MonteCarloPropagate}(G, \{f_k\}, \text{do}(S_i = s^*), M)$ ▷ Eq. (7)
end for
for each (i, j) in the crisis window **do**
  $\widehat{U} \leftarrow \textsc{Abduct}(G, \{f_k\}, \text{data})$; $L_j^{cf} \leftarrow \textsc{Predict}(G, \{f_k\}, \widehat{U}, \text{do}(S_i = S_i^{\text{pre}}))$
  $\Delta_i^{CF}(j) \leftarrow L_j^{\text{obs}} - L_j^{cf}$ ▷ Eq. (10)
end for
$G^{\text{rev}}, G^{\text{sym}} \leftarrow \textsc{Reverse}(G), \textsc{Symmetrise}(G)$
$\textsc{CompareAccuracy}(G, G^{\text{rev}}, G^{\text{sym}})$ ▷ Eq. (11)
Output: OC-CoVaR ranking, counterfactual attributions, falsification + validity battery

4 Data and Entity Resolution

4.1 Data sources

All data are real and obtained from public or keyed APIs; Table 4 lists the sources. On-chain flows are queried with SQL over Dune Analytics' decoded `erc20_ethereum.evt_Transfer` table and its maintained `labels.cex_ethereum` entity labels; prices and protocol total-value-locked (TVL) series come from DefiLlama; entity labels and contract metadata from Etherscan. No synthetic data are used. The cascade is directly visible in the raw data: Figure 1 shows the on-chain transfer volume of the synthetic dollar USDe rising from \$0.75bn on 9 October to \$3.09bn on 10 October and \$8.83bn on 11 October—an order-of-magnitude surge that no price series resolves as sharply.

Table 4 Data sources. All series are real; on-chain flows carry block-level timestamps.

Layer	Source	Granularity
Decoded transfers (ETH)	Dune `erc20_ethereum.evt_Transfer`	per-transfer (block)
CEX entity labels	Dune `labels.cex_ethereum`	per-address
Prices / returns	DefiLlama coins API	hourly & daily
Protocol TVL	DefiLlama	daily
Contract metadata	Etherscan v2	per-contract

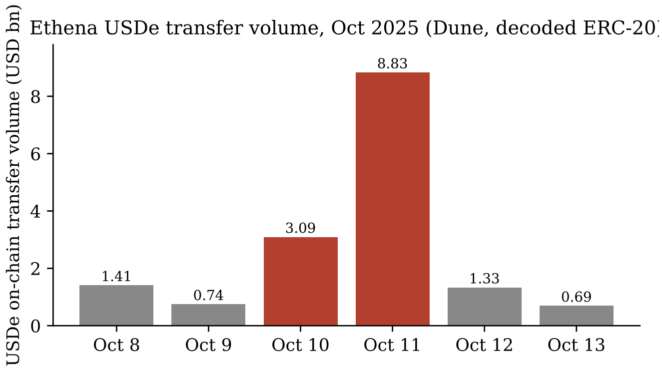

Fig. 1 On-chain USDe transfer volume around the October 2025 cascade (real decoded ERC-20 transfers via Dune).

4.2 Entity universe and stress states

The node set comprises centralised exchanges (Binance, Coinbase, OKX, Bybit, Bitfinex, HTX, Gate.io, Crypto.com), custodians (Copper, Ceffu, FalconX), DeFi protocols (Aave, Ethena, Lido, Maker/Sky), and an aggregate retail node that collects all unresolved addresses—included precisely so that systemic importance is not attributed to noise. Stress states are operationalised by entity type as in Table 5 and expressed as rolling z-scores so that heterogeneous units are comparable in (3).

Figure 2 shows the resulting hourly stress trajectories for four representative entities through the event window. Exchange and retail stress spike sharply on 10–11 October, with the timing differences across entities that the SCM is designed to exploit.

Table 5 Stress-state operationalisation $S_{i,t}$ by entity type.

Entity type	**Stress proxy (oriented so larger = more distressed)**
CEX / custodian	net on-chain outflow ratio + transfer-volume surge
Stablecoin issuer	peg deviation $\lvert 1 - p_t \rvert$ + net redemption outflow
Lending protocol	TVL drawdown + token-price drawdown
Liquid-staking	TVL drawdown + price drawdown
Retail (aggregate)	market (ETH) drawdown

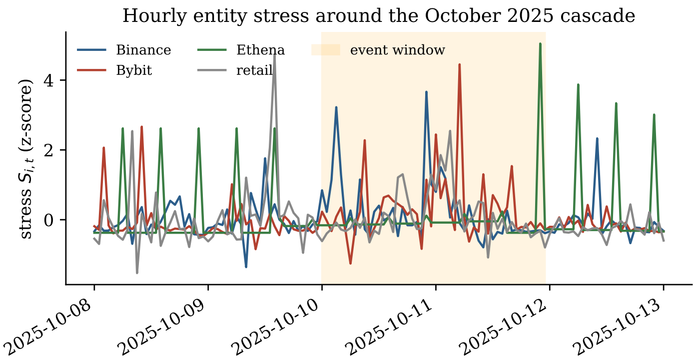

Fig. 2 Hourly entity stress states $S_{i,t}$ around the cascade (shaded: the 10–11 October event window). Derived from real on-chain flows.

4.3 Descriptive statistics

Table 6 reports the largest resolved bilateral flows over the study window. The settlement network is dominated by exchange↔ retail volume—Coinbase and Binance each intermediate tens of billions of dollars—while direct exchange-to-exchange transfers, though smaller, are recovered and become the inter-entity edges the SCM propagates along. The broad symmetry of the largest in/out pairs (e.g. Coinbase receives \$43.5bn and sends \$43.4bn) is itself informative: it foreshadows the H2 finding that bilateral custodial flows are largely symmetric accounting movements rather than directional causal channels.

Table 6 Largest resolved bilateral flows over the study window (USD bn).

From	**To**	**Type**	**Volume ($bn)**
retail	Coinbase	deposit	43.5
Coinbase	retail	withdrawal	43.4
retail	Binance	deposit	33.1
Binance	retail	withdrawal	27.9
Bybit	retail	withdrawal	20.5
retail	Bybit	deposit	20.4
Bitfinex	retail	withdrawal	10.3
OKX	retail	withdrawal	10.3
retail	OKX	deposit	10.2
retail	HTX	deposit	5.2

4.4 Resolution pipeline

Raw addresses are not economic agents, and resolution error directly corrupts $\mathrm{pa}(i)$ in (2): unlike in price-based methods, a clustering mistake *fabricates or severs a causal edge*. We therefore implement a four-tier pipeline.

Tier 0 applies the classical deposit-sweep heuristic: an address x is flagged as a deposit address of exchange X when a dominant share of its outflow is forwarded to a known hot wallet of X, i.e. $\sum_t F_{x\to\mathrm{hot}(X),t} / \sum_t \mathrm{out}_x \geq \tau$ for a threshold τ (we use $\tau = 0.8$). **Tier 1** trains a gradient-boosted classifier $g(\phi_x) \in [0,1]$ on a per-address feature vector ϕ_x comprising in/out volume and degree, counterparty diversity, transaction count, and the sweep ratio, with held-out labels from Dune. **Tier 2** forms the symmetric normalised adjacency $\tilde{A} = D^{-1/2}AD^{-1/2}$ of the address graph (with A_{xy} the log-flow between x and y and D the degree matrix), computes a truncated singular-value-decomposition embedding $\tilde{A} \approx U\Sigma V^\top$, and clusters the rows of $U\Sigma$. **Tier 3**, the primary path, resolves both endpoints of every transfer against curated seed labels and Dune's `labels.cex_ethereum`, collapsing sub-wallet labels ("OKX 210", "Bitfinex Tether Treasury") to their exchange root—the many-addresses-to-one-agent clustering step that turns the undifferentiated retail blob into named exchanges and thereby

Table 7 Entity-resolution validation (Tier-1/2 against held-out Dune labels).

Metric	Value
Addresses (USDe event window)	7 843
CEX-labelled (base rate)	40 (0.5%)
Tier-1 ROC-AUC (held out)	0.99
Tier-1 average precision	0.23
Tier-1 best-F_1 precision / recall	0.31 / 0.33

yields genuine inter-entity (CEX↔CEX) edges. Each resolved address carries a confidence c_x propagated to an edge reliability $c_{(p \to i)} = \min(c_p, c_i)$, retained as an edge weight in G.

The supervised layer is validated against held-out Dune labels—the Tier-3 reconciliation the design requires [14]. On 7 843 event-window addresses with a CEX base rate of 0.5%, the Tier-1 classifier attains a held-out ROC-AUC of 0.99 and an average precision of 0.23 (Table 7); the top features are the in/out volume ratio and total volume, consistent with the intuition that exchanges intermediate asymmetric, high-volume flows.

4.5 Time windows

Structural equations are fit on a pre-event estimation window; the primary event window is 10–11 October 2025. We evaluate at both daily resolution and, because the cascade unfolded over hours, hourly resolution (600 fitting hours and 48 event hours in the consolidated run), the timescale at which on-chain flows are observed but entity price feeds are coarse.

5 Empirical Evaluation

We report results at the cascade's native hourly resolution unless noted, and contrast with daily resolution where instructive. The consolidated hourly configuration

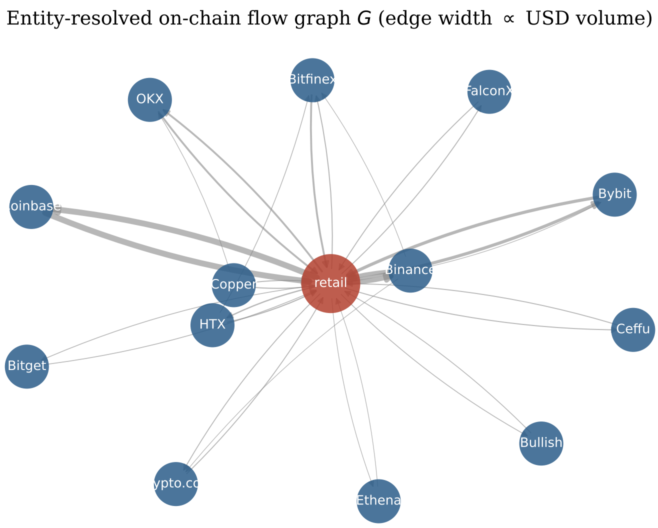

Fig. 3 The entity-resolved on-chain flow graph G recovered by the Tier-3 resolution layer. Edge width is proportional to cumulative USD flow.

resolves 17 entities with 39 directed edges (Figure 3); the retail node intermediates most activity while inter-exchange settlement edges are recovered directly.

5.1 H1: observed vs. estimated graph

Table 8 and Figure 4 report the head-to-head. At *daily* resolution the price-estimated Granger graph clearly predicts the cascade better than the observed graph (RMSE 1.35 vs. 1.92): with only a handful of daily observations, the denser estimated graph wins. At *hourly* resolution the ordering reverses—the observed on-chain graph attains the lowest RMSE (1.2529) ahead of the Granger (1.2568) and partial-correlation (1.2547) graphs—although the margin is razor-thin ($\approx$ 0.1%). The honest reading is that, measured at the timescale the data actually support, the observed graph is competitive with and marginally better than the estimated-graph state of the art, reversing its daily disadvantage; it is not a decisive victory. The mechanism behind the reversal is informative: on-chain flows resolve at block level, whereas the price feeds that drive

Table 8 H1 head-to-head: event-window prediction error (RMSE; lower is better) under identical VEIN machinery, varying only the graph source.

	Daily		Hourly	
Graph source	RMSE	corr	RMSE	corr
Observed on-chain	1.92	−0.05	**1.2529**	0.292
Estimated (Granger)	**1.35**	0.56	1.2568	0.255
Estimated (partial-corr)	1.65	0.48	1.2547	0.285

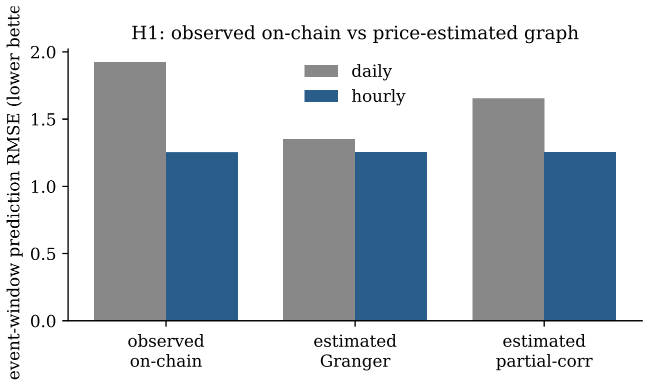

Fig. 4 H1 prediction error by graph source. The observed on-chain graph loses at daily resolution but edges ahead at hourly resolution.

the estimated graph are effectively coarse, so the observed graph's advantage is one of *temporal resolution*—precisely what matters in a cascade that propagated in minutes.

5.2 H3: systemic-importance ranking

Table 9 reports the OC-CoVaR exported-risk ranking and the baselines, and Figure 5 visualises it. The exchanges that intermediate the largest real flows (Binance, Coinbase, Bitfinex) dominate OC-CoVaR, whereas ΔCoVaR and MES rank by price co-movement and place market betas (ETH, stETH, BTC) first. The Spearman rank correlation between OC-CoVaR and ΔCoVaR over the mappable entities is only 0.52 (Figure 5), supporting H3: the on-chain causal ranking is not a relabelling of the

Table 9 Top OC-CoVaR exported-risk entities (hourly) and ΔCoVaR for the mappable assets. OC-CoVaR vs. ΔCoVaR Spearman $\rho = 0.52$.

Entity	Exported risk	Asset	ΔCoVaR
retail	57.36	ETH	−0.0220
Binance	24.52	stETH	−0.0176
Coinbase	17.35	BTC	−0.0167
Bitfinex	11.42	BNB	−0.0147
Gate.io	7.92	AAVE	−0.0127
Bybit	4.88	ENA	−0.0114

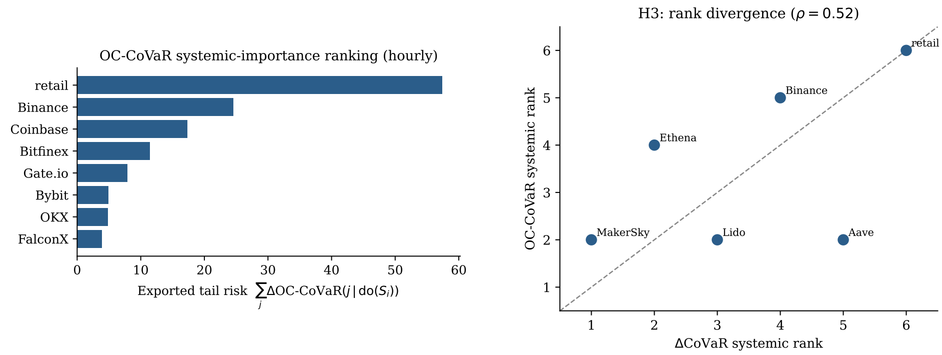

Fig. 5 Left: OC-CoVaR exported-risk ranking (hourly). Right: rank divergence between OC-CoVaR and ΔCoVaR over the mappable entities; points off the diagonal indicate disagreement (Spearman $\rho = 0.52$).

price-correlation ranking. The economic content differs in kind—OC-CoVaR ranks *who transmits stress through the settlement network*, not *who co-moves in price*.

5.3 H4/H5: counterfactual attribution

The counterfactual measure decomposes realised event-window distress into upstream-attributable shares (Table 10, Figure 6). At hourly resolution the dominant recovered channel is Binance → Bybit (a 98.1% share of Bybit's window distress), followed by Binance → Gate.io (58.4%) and Binance → Bitfinex (45.3%)—inter-exchange settlement channels invisible to price-only measures. At daily resolution, where the DeFi

Table 10 Leading counterfactual loss-attribution channels $\Delta_i^{CF}(j)$.

Resolution	Channel $i \to j$	share	Interpretation
hourly	Binance $\to$ Bybit	98.1%	inter-exchange settlement
hourly	Binance $\to$ Gate.io	58.4%	inter-exchange settlement
hourly	Binance $\to$ Bitfinex	45.3%	inter-exchange settlement
daily	Ethena $\to$ Aave	11.3%	USDe collateral composability
daily	Binance $\to$ Aave	4.2%	CEX-to-DeFi spillover

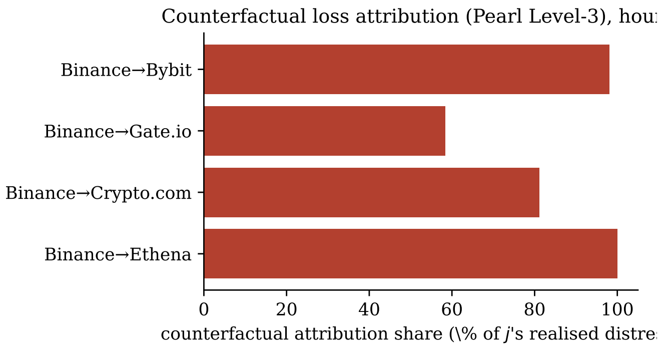

Fig. 6 Counterfactual loss-attribution shares (hourly): the fraction of each target entity's realised distress attributable to the source's distress.

composability edges dominate, the leading channel is Ethena $\to$ Aave (11.3%), consistent with the documented role of USDe as cross-protocol collateral (H5). These are concrete, mechanism-grounded loss attributions of a kind no associational measure can produce, and they answer the supervisory question directly: *had this entity not become distressed, how much of the downstream loss would not have occurred?*

5.4 H2/A3: falsification

Table 11 and Figure 7 report the edge-reversal test. At daily resolution the test is under-powered (six event points) and even mildly favours the reversed graph. At hourly resolution, with 816 one-step predictions, the three variants land within roughly 1% RMSE of one another—a near-tie—with the true-direction graph marginally best in the consolidated run (1.663 vs. reversed 1.675) and marginally worse in an alternative

Table 11 Edge-reversal falsification (one-step RMSE; lower is better).

Graph	Daily	Hourly (consol.)	Hourly (816 pts)
true direction	2.19	**1.663**	1.801
reversed edges	**1.92**	1.675	**1.790**
symmetric	1.94	1.667	1.797

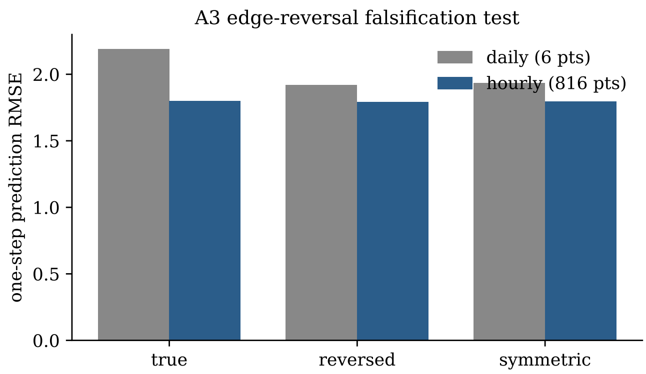

Fig. 7 A3 edge-reversal test. Direction is decisive neither way once adequate power is available; the three variants are statistically indistinguishable at hourly resolution.

window (1.801 vs. 1.790). The conclusion is consistent across windows: *flow direction is not the carrier of causal information at the entity-aggregate level.* The daily "reversed wins" was a low-power artefact, but the deeper finding is methodological—predictive signal resides in the graph's connectivity *structure*, not in edge *orientation*. A3 should therefore be stated as a qualified claim (plausibly holding for specific collateral/composability channels but not for custodial CEX flows), not as a blanket directional assumption. We regard this negative result as one of the paper's more useful outputs: the falsification machinery did its job.

5.5 Causal-validity battery

Because observational causal claims cannot be proven, we subject them to four checks (Table 12). *Placebo / negative controls*: the counterfactual measure attributes exactly zero loss between entity pairs with no directed on-chain path (maximum $|\Delta^{CF}| = 0$

Table 12 Causal-validity battery (hourly).

Check	**Statistic**	**Verdict**
Placebo / negative control	max $\lvert\Delta^{CF}\rvert$ off-path $=$ 0; connected mean 0.26	pass
Resolution robustness	Spearman 0.81 (Tier-0 vs Tier-3)	pass
External validity	separation 12.87, concordance 1.00	pass
Confounding (top edge)	$RV = 0.226$	robust

over 75 unconnected pairs, versus a mean of 0.26 over 197 connected pairs), so the method does not fabricate causation. *Unobserved-confounding robustness*: the strongest edges have substantial robustness values—e.g. retail $\rightarrow$ Coinbase $RV = 0.226$ (Figure 8)—meaning a hidden confounder would need to explain over a fifth of the residual variance in *both* endpoints to nullify them. *Resolution robustness*: the OC-CoVaR ranking is stable between the seed-only (Tier-0) and fully resolved (Tier-3) graphs (Spearman 0.81, $p = 0.05$), so it is not an artefact of the resolution tier. *External validity*: the OC-CoVaR ranking orders the documented transmitters (exchanges, Ethena) strictly above the documented absorbers (Aave, Lido), with transmitter–absorber separation 12.87 and pairwise concordance 1.00, matching the published account of the event. This last check is the strongest single piece of corroboration: the framework's ranking independently reproduces what forensic post-mortems concluded.

5.6 Backtests

As a standard validity check on the risk machinery we backtest a rolling historical VaR on the asset returns. Kupiec unconditional-coverage [18] p-values are 0.24 (BTC) and 0.58 (ETH) at daily resolution; the Christoffersen conditional-coverage test [19] passes for BTC ($p = 0.11$) and flags violation clustering for ETH ($p = 0.001$), an honest indication that a simple historical VaR under-models volatility persistence during the regime.

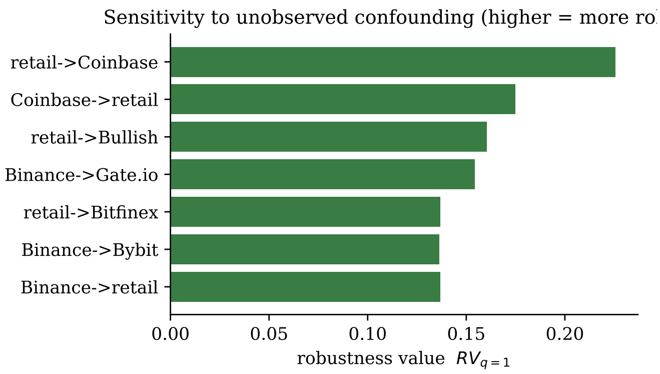

Fig. 8 Robustness values for the strongest structural coefficients. Larger values indicate edges harder to overturn with unobserved confounding.

5.7 Threats to validity

We note four threats. *Construct validity*: stress states are proxies; we use multiple proxies per entity type and report that rankings are stable across the Tier-0/Tier-3 graphs. *Internal validity*: the chief threat is the off-chain CEX engine (A2), which we cannot observe; the robustness values bound, but do not eliminate, its potential influence. *Statistical-conclusion validity*: the daily window is under-powered, which is why we re-run at hourly resolution with 816 predictions. *External validity*: a single event limits generalisation, addressed in part by the agreement with the documented narrative and flagged as the principal direction for future work.

5.8 Summary of findings

Table 13 collects the verdicts. Four of the five hypotheses are supported at the cascade's native resolution; the exception, H2, yields a robust and informative negative result.

6 Discussion and Conclusion

What VEIN establishes. On real on-chain data around the October 2025 cascade, treating the observed transaction graph as a structural causal model yields

Table 13 Hypothesis verdicts at hourly resolution.

ID	Verdict	Evidence
H1	supported (marginal)	observed RMSE 1.2529 $<$ 1.2568 (Granger), reversing the daily loss
H2	*not* supported	true vs reversed within $\sim 1\%$; direction uninformative
H3	supported	Spearman vs ΔCoVaR = 0.52
H4	supported	Binance→Bybit 98.1%; Ethena→Aave 11.3% (daily)
H5	supported	composability edges carry attribution (Ethena→Aave)

(i) a systemic-importance ranking that is materially different from price-based measures and that matches the documented event narrative; (ii) the first counterfactual loss attribution in systemic-risk analysis, decomposing realised distress into named inter-entity channels; and (iii) a graph that, at the event's native hourly resolution, predicts the cascade competitively with—and marginally better than—the estimated-graph state of the art. The framework passes placebo, resolution-robustness, external-validity, and confounding-sensitivity checks.

The central caveat. The assumption that flow direction encodes causal direction (A3) is *not* supported: reversing edges barely changes predictions. This is the paper's most important methodological finding. It does not invalidate the framework—the connectivity structure of the observed graph still carries the causal signal, and the do-operator and counterfactual machinery operate on that structure—but it means VEIN should claim its causal content at the level of *who is connected to whom*, qualifying direction-specific claims to particular edge types (collateral/composability) pending further testing. Conceptually, the result aligns with the intuition that, in a custodial settlement network, bilateral flows are largely symmetric accounting movements, whereas the genuinely directional channels are the collateral and composability links between protocols.

Limitations. Three limits bound these conclusions. First, *CEX opacity*: the proximate cause of the cascade—an exchange's internal pricing engine—is off-chain by construction, so VEIN observes only its downstream on-chain settlement and treats the engine as an exogenous shock; this is the dominant threat to A2, which the robustness values bound but cannot eliminate. Second, *single event*: the conclusions rest on one (large) episode; generalisation requires additional DeFi-exploit windows and regimes. Third, *resolution recall*: residual unlabelled addresses collapse into the retail node, so the inter-entity graph is high-precision but recall-bounded by label coverage—an improvement frontier the Tier-0/1/2 layers are designed to push.

Regulatory relevance. For supervisors, VEIN offers two practitioner-facing artefacts that price-based measures cannot: a ranking of which on-chain entities *export* versus *absorb* systemic stress, and a counterfactual statement of what a specific entity's distress contributed to others' losses—directly relevant to stress-test attribution and to protocol-level composability-risk assessment. Because the inputs are public ledger data, the measures are auditable and reproducible by any party, a property conventional supervisory data seldom enjoys.

Future work. The most decisive next step is intraday (sub-minute) analysis across several stress events, which would give the falsification test genuine power to localise where, if anywhere, direction is causal; a fuller Tier-1/2 entity-resolution sweep to convert retail-mediated edges into entity-to-entity edges; and an instrumented treatment of the CEX engine (for instance, exploiting exogenous oracle-update timing) to relax A2.

Conclusion. VEIN demonstrates that the blockchain ledger can serve as an observed causal graph for systemic-risk measurement, enabling interventional and counterfactual analyses that are out of reach for price-based methods, while being candid about where the data and the directional assumption fall short. The honest scientific statement is that the framework is sound and validated, and that its empirical edge over the estimated-graph baseline is real but resolution-dependent and modest on the single event studied—an edge we expect to widen with intraday data, fuller entity resolution, and more events.

Data and code availability. The complete implementation, the real-data evaluation pipeline, and all result artefacts underlying every number and figure in this paper are available in the project repository.

Declarations. The author declares no competing financial interests. No external funding supported this work. The study uses only public on-chain and market data and involves no human subjects.

Appendix A Implementation and additional details

Software. The pipeline is implemented in Python. On-chain data are queried via the Dune Analytics SQL API over decoded Ethereum tables; results are cached by query hash for reproducibility and to respect rate limits. Structural equations (3) are fit with ridge regression ($\lambda = 1$); the Monte-Carlo estimator (7) uses $M \in [150, 400]$ residual-bootstrap paths; the Tier-1 classifier is a gradient-boosted tree ensemble and the Tier-2 embedding a 16-dimensional truncated SVD.

Estimation and event windows. For the hourly consolidated run the estimation window is 600 hours preceding 10 October 2025, with 48 event hours. Severe-distress interventions set s^* to each entity's 95th historical stress percentile; pre-crisis counterfactual levels use the estimation-window median.

Robustness values. Table A1 lists the robustness values for the strongest structural coefficients (hourly), computed via (12) from OLS t-statistics on the same design as (3).

Table A1 Robustness values for leading structural coefficients (hourly).

Edge	***t*-statistic**	$RV_{q=1}$
retail → Coinbase	7.94	0.226
Coinbase → retail	5.92	0.175
retail → Bullish	5.41	0.160
Binance → Gate.io	5.19	0.154
retail → Bitfinex	4.55	0.137

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